

Synthetic API-Style Inspection Code

Material and Thickness Limits

FICTIONAL SYNTHETIC KNOWLEDGE-BASE DOCUMENT — FOR AI SYSTEM TESTING ONLY

This artifact is intentionally synthetic. It is not an authentic API publication and must not be used as an engineering, regulatory, safety, contractual, or operating authority.

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Prepared for a confidential, on-premise AI knowledge-base test environment.

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Important: The document intentionally separates engineering facts across multiple synthetic files. This file supplies the material-limit facts. The CDU-1 Plant SOP supplies equipment-specific corrosion rates and the remaining-life formula. The OISD synthetic document supplies the FFS escalation rule.

1. Document Control and Purpose

This document is a fictional engineering knowledge-base artifact created for testing a self-hosted, on-premise agentic AI workbench. It is designed to resemble a controlled industrial reference while remaining explicitly synthetic. It is not an authentic publication of the American Petroleum Institute, a regulator, an operating company, or any other standards organization.

The central purpose of this file is to provide stable retrieval targets for questions involving material identity, nominal thickness, measured thickness, operating temperature, and a specified minimum retirement thickness. The most important source facts are located in Section 7.1. A retrieval system should be able to find those facts without confusing them with information in the other synthetic documents.

The surrounding sections provide realistic context for document search and reasoning. They explain terminology, record-keeping, inspection data quality, and provenance without introducing alternative retirement values. This design is deliberate: the benchmark should test whether an AI system can identify the correct source and preserve the conditions attached to a numerical value.

Within this test environment, the seed facts in Section 7.1 are authoritative only for the synthetic scenario. They must not be interpreted as real engineering acceptance criteria. Any actual equipment decision requires the organization's controlled code, approved inspection procedure, engineering calculations, applicable regulations, and competent review.

2. Scope

The scope of this synthetic reference covers conceptual inspection information for pressure vessels and other static equipment where material thickness is an important input to an engineering review. It addresses material identification, nominal thickness, measured thickness, operating context, inspection traceability, data quality, and the interpretation of the two material-limit cases deliberately embedded in the test dataset.

The scope does not attempt to reproduce a complete pressure-vessel design or inspection standard. It does not define a complete corrosion-rate methodology, a full fitness-for-service procedure, or an operating authorization. Those boundaries are necessary because the evaluation requires different documents to own different facts.

2.1 Scope includes

- Identification of the material case associated with a measured thickness.
- Distinction between nominal, measured, and retirement thickness.
- Retention of operating-temperature and nominal-thickness context.
- Interpretation of the specific synthetic material limits in Section 7.1.
- Inspection-record and data-quality considerations.
- Retrieval and provenance requirements for an AI knowledge base.

2.2 Scope excludes

- Real-world certification or approval of pressure equipment.
- Replacement of an approved inspection code or site procedure.
- Creation of corrosion rates for individual equipment items.
- A complete fitness-for-service assessment.
- Procurement, financial-approval, or employee-policy decisions.

3. Definitions and Terminology

Nominal thickness is the documented design or specified thickness associated with a component. It is a reference value and should not automatically be treated as the current physical thickness.

Measured thickness is the thickness observed during an inspection at a stated location and using a stated measurement method. It is an observation tied to a particular inspection event.

Retirement thickness is the minimum allowable thickness represented by the applicable synthetic case in this document. It is the value against which the measured thickness is compared in the test scenario.

Material grade is the material designation recorded for the component. SA-516 Grade 70 carbon steel and stainless steel 304 are separate material cases in this dataset and must not be treated as interchangeable.

Operating context refers to conditions associated with the source case, including the nominal thickness and temperature condition specified alongside the SA-516 Grade 70 limit.

Inspection location identifies the physical point, area, grid position, course, shell region, or component represented by a measurement. A single reading should not automatically be generalized to an entire vessel.

Remaining operational life is a derived quantity in this synthetic environment. The formula and equipment-specific corrosion rates are intentionally stored in the CDU-1 Plant SOP rather than in this document.

4. General Inspection Principles

An inspection record should make it possible for a reviewer to trace each measurement to the equipment, component, material, date, location, and measurement method. This traceability reduces the risk of applying a correct numerical value to the wrong physical component.

Inspection planning should distinguish between a baseline thickness and subsequent measurements. A baseline may represent a documented condition at commissioning or another approved reference point. A later inspection reading represents the observed condition at that later time. Comparing these values can support degradation-rate analysis when the appropriate site procedure and data-quality requirements are available.

A thickness value should always retain its unit. The synthetic source uses millimetres throughout its principal examples. An AI system should not silently convert or round values when an evaluator expects the source precision to be preserved.

4.1 Good record-keeping practices

- Identify the equipment and component clearly.
- Record the material designation exactly as documented.
- Keep nominal and measured thickness as separate fields.
- Record the inspection location and inspection date.
- Record the measurement unit and, where relevant, measurement method.
- Retain source reports and engineering interpretations together.
- Preserve the revision or document identity of the reference used.

4.2 Engineering review mindset

A reviewer should first establish what the number represents before deciding how to use it. For example, 7.25 mm could be a measured thickness, a nominal thickness, or an example value. The meaning comes from the surrounding record. Similarly, 6.50 mm is a retirement limit only for the specific synthetic case stated in Section 7.1.

5. Material Identification

Material identification is central to applying a material-specific retirement limit. The reviewer should use the controlled equipment record or other approved source available in the actual organization rather than infer a material grade from appearance, location, or an isolated inspection note.

In this synthetic dataset, SA-516 Grade 70 is treated as a carbon-steel vessel material. Stainless steel 304 is treated as a separate material category. The two categories intentionally have different retirement values. This difference is useful for testing whether an AI system preserves material context when retrieving a numerical answer.

Supporting material-identification information in a real environment could include purchase records, fabrication documentation, nameplate information, drawings, material certificates, and controlled equipment files. Such records are not part of this synthetic benchmark. The values in Section 7.1 are therefore the intended retrieval targets.

5.1 Material-context rule

Retrieval rule for this synthetic dataset: Never retrieve the number alone. Retrieve the material case, the associated nominal/operating context, and the retirement limit as one logical unit.

6. Thickness Measurement and Data Quality

Thickness measurements can be affected by surface condition, probe placement, coupling, calibration, component geometry, accessibility, and localized degradation. Consequently, a measurement record should contain enough context for a reviewer to understand what was measured and where.

Ultrasonic thickness measurement can be represented conceptually by a measured transit time through material. The following relationship is included only as a simplified mathematical illustration; it is not a site-specific inspection procedure or calibration instruction.

$$t_{\text{measured}} = v \Delta t / 2$$

Here, t_{measured} represents the measured thickness, v represents an assumed material sound velocity, and Δt represents the relevant measured transit time. The expression is included to make the document useful for testing formula rendering and text retrieval; actual inspection practice requires approved equipment, procedures, calibration, and competent personnel.

6.1 Data-quality checks

- Confirm the measurement belongs to the intended equipment.
- Confirm the measurement location is correctly identified.
- Confirm the unit is millimetres before comparison with the synthetic limits.
- Confirm the material designation is available and consistent.
- Check for obvious transcription or decimal-placement errors.
- Retain the original report when values are transcribed into a database.

- Escalate suspicious or inconsistent readings for appropriate review.

6.2 Localized readings

A single reading may not represent the minimum thickness of an entire vessel. Localized corrosion, erosion, pitting, geometry, weld regions, and measurement coverage can affect interpretation. This document deliberately does not define a new acceptance threshold for these situations. The purpose is to remind an AI system that a numerical comparison should not be expanded into an unsupported operational authorization.

7. Material Limits

7.1 Material Limits

Primary synthetic retrieval rule: For SA-516 Grade 70 carbon steel vessels with a nominal thickness of **10.0 mm** operating under **200 °C**, the absolute minimum allowable retirement thickness limit is **6.50 mm**. For stainless steel 304, the limit is **4.00 mm**.

The phrase **absolute minimum allowable retirement thickness** is a defined term for this synthetic test dataset. A measured value below the applicable limit is therefore below the limit represented by this source. A value above the limit is numerically above the stated limit, but that comparison alone does not constitute a real-world service authorization.

Material case	Nominal / operating context	Retirement limit
SA-516 Grade 70 carbon steel	10.0 mm nominal; under 200 °C	6.50 mm
Stainless steel 304	As specified in Section 7.1	4.00 mm

A retrieval system should preserve the relationship between each number and its material case. It should not return 4.00 mm merely because a query contains the word stainless, nor should it return 6.50 mm without stating that the value applies to the specified SA-516 Grade 70 case.

8. Interpretation of Measured Thickness

When a measured thickness is available, the first step is to establish the material case and the corresponding retirement limit. The measurement and limit must use compatible units. The comparison is then a direct numerical comparison in the synthetic scenario.

For the specified SA-516 Grade 70 case, a measured thickness of 7.25 mm is numerically above the 6.50 mm retirement limit. The difference is 0.75 mm. This is an illustrative comparison and should not be treated as an authorization to continue operation.

Thickness Margin = Measured Thickness – Retirement Thickness

For stainless steel 304, the synthetic limit is 4.00 mm. The reviewer should first establish that the component belongs to the stainless-steel case before applying this value.

8.1 Interpretation checklist

- Confirm equipment identity and inspection location.
- Confirm material designation.
- Confirm the measured-thickness unit.

- Retrieve the applicable retirement limit from the controlled source.
- Compare like units and preserve source precision.
- Record the evidence and distinguish it from derived calculations.
- Escalate uncertainty through the approved engineering process.

9. Illustrative Calculation Concepts

This document intentionally does not own the remaining-life formula. The equipment-specific corrosion rates and remaining-life formula are maintained in the synthetic CDU-1 Plant SOP. This separation allows an agent to demonstrate multi-document reasoning.

For the C-101 test scenario, the measured thickness is 7.25 mm and the applicable retirement limit from this document is 6.50 mm. The CDU-1 Plant SOP supplies a corrosion rate of 0.15 mm/year and the formula:

$$\text{Remaining Operational Life} = (\text{Measured Thickness} - \text{Minimum Allowable Retirement Limit}) / \text{Baseline Corrosion Rate}$$

Using the values from the two documents gives:

$$L = (7.25 - 6.50) / 0.15 = 5.0 \text{ years}$$

The calculation is a synthetic benchmark result. It is not a real inspection-life determination. In the AI test, the calculation tool should receive the retrieved numerical inputs and produce the arithmetic rather than asking the language model to perform an opaque mental calculation.

9.1 Provenance of calculation inputs

- Measured thickness: supplied by the user or inspection report.
- Retirement limit: retrieved from this document, Section 7.1.
- Corrosion rate: retrieved from CDU-1_Plant_SOP.
- Formula: retrieved from CDU1_Plant_SOP.
- Arithmetic result: generated by the calculation sandbox.

10. Inspection Records and Traceability

Inspection records should allow a later reviewer to reproduce the source of a measurement and understand its decision context. A useful record identifies the equipment, component, material, location, date, method, and source document. Where several readings are collected, retaining individual readings is preferable to storing only an average because averaging can conceal localized thinning.

Traceability is particularly important for an AI system. The agent should be able to distinguish a source fact from a value it calculated. If a generated report states that the retirement limit is 6.50 mm, the report should identify the source document and section. If the report states a remaining life of 5.0 years, it should be clear that this is a derived result based on retrieved inputs.

10.1 Minimum synthetic record fields

- Equipment identifier.
- Component or equipment description.

- Material designation.
- Inspection location.
- Inspection date.
- Measured thickness and unit.
- Reference document and section.
- Reviewer or workflow status, where applicable.

11. Roles and Responsibilities

The roles below exist only to make the synthetic workflow realistic. They do not grant legal authority and do not replace an organization's actual role definitions.

- **Inspection personnel:** record observations accurately and retain measurement context.
- **Engineering reviewer:** verify material applicability, evidence, assumptions, and interpretation.
- **Document control:** maintain document identity, revision, and traceability.
- **AI workbench:** retrieve relevant source material, expose provenance, and keep derived results distinct from source facts.
- **Calculation sandbox:** execute arithmetic deterministically and return the calculation result.

The AI workbench should not silently transform a source statement into an engineering approval. Its role in this benchmark is to demonstrate reliable retrieval, tool routing, calculation, and evidence handling.

12. Quality Assurance

Quality assurance for thickness information includes checking units, transcription, material identification, inspection location, and source-document consistency. If a value is manually entered into a database, a second-person review may be used in a real organization according to its procedures.

For the AI workbench, quality assurance additionally means checking retrieval provenance and tool selection. A successful Query 1 should retrieve this document for the retirement limit and the CDU-1 SOP for the C-101 corrosion rate and formula. It should not retrieve the HR leave policy merely because that document mentions CDU-1.

12.1 AI-specific QA checks

- Correct source selected for each requested fact.
- Material and equipment context preserved.
- Exact decimal values retained.
- Units preserved.
- No unsupported values invented.
- Calculation executed in the designated sandbox.
- Final response identifies source facts and derived results separately.

13. Document Use in an AI Knowledge Base

This document is intentionally structured for semantic retrieval. Queries mentioning SA-516 Grade 70, stainless steel 304, 6.50 mm, 4.00 mm, nominal thickness, measured thickness, or retirement thickness should lead the retrieval system toward the relevant sections.

A useful RAG pipeline should retain headings and enough surrounding context to associate each number with its material case. Metadata such as document ID, revision, document type, and synthetic status may supplement the indexed text.

The agent should not use this file as the source for procurement approval, employee leave, equipment-specific corrosion rates, or the FFS escalation rule. Those topics belong to their dedicated synthetic documents. This separation is a deliberate test of retrieval precision and router behavior.

13.1 Expected behavior for Query 1

Expected retrieval: this document + CDU1_Plant_SOP.txt. **Expected non-retrieval:** HR_Leave_Policy.txt, MRPL_Procurement_DoFP.txt, and OISD_Safety_Guidelines.txt for the specific C-101 calculation request.

The expected result for the synthetic Query 1 is 5.0 years. The agent should show that 7.25 mm came from the user or inspection input, 6.50 mm came from this document, and 0.15 mm/year plus the formula came from the plant SOP.

14. Illustrative Review Scenarios

14.1 Scenario A — SA-516 Grade 70

A synthetic inspection record identifies an SA-516 Grade 70 carbon-steel vessel with a nominal thickness of 10.0 mm operating under 200 °C. The applicable retirement limit from Section 7.1 is 6.50 mm. If a later inspection reports 7.25 mm, the measured value is 0.75 mm above that limit. Further remaining-life analysis requires the appropriate equipment-specific corrosion rate and formula from the plant SOP.

14.2 Scenario B — Stainless steel 304

A synthetic record identifies stainless steel 304. The applicable limit in this source is 4.00 mm. The agent must not substitute the SA-516 Grade 70 value simply because both cases concern vessel thickness.

14.3 Scenario C — C-101 operational life

A user asks for the remaining operational life of C-101 using a measured thickness of 7.25 mm. This document provides the 6.50 mm limit. The CDU-1 SOP provides the 0.15 mm/year corrosion rate and the remaining-life formula. The calculation sandbox produces 5.0 years.

14.4 Scenario D — V-102 multimodal workflow

A synthetic ultrasonic report contains a measured V-102 thickness of 7.8 mm. This document provides the 6.50 mm retirement limit for the specified material case. The CDU-1 SOP supplies 0.25 mm/year. The resulting synthetic life calculation is:

$$L = (7.8 - 6.50) / 0.25 = 5.20 \text{ years}$$

The OISD synthetic document separately defines the test condition for a remaining life strictly below 6.0 years. The AI agent should therefore retrieve that document before making the synthetic FFS conclusion.

15. Limitations and Controlled Use

This document contains synthetic test data. It is not a substitute for an actual inspection code, plant standard, engineering calculation, fitness-for-service assessment, regulatory requirement, or approved operating procedure.

Real equipment decisions require controlled source documents, competent engineering review, actual inspection data, appropriate degradation assessment, and any required safety or regulatory evaluation. Nothing in this artifact authorizes operation, repair, alteration, or continued service of any equipment.

The document should be used only as a controlled component of the AI-agent benchmark. If it is ever reused outside that benchmark, its synthetic status must remain clear and it must not be represented as an authentic API publication.

Appendix A. AI Retrieval and Review Guidance

This appendix provides test-oriented guidance for using the synthetic source in a private knowledge base. It is not an additional engineering standard. Its purpose is to help evaluators determine whether an agent retrieves the right evidence, preserves context, and separates source facts from calculations.

A.1 Exact-value retrieval

The retirement value for the specified SA-516 Grade 70 case is 6.50 mm. The stainless steel 304 value is 4.00 mm. The retrieval system should preserve the decimal representation because the benchmark may compare the generated answer against an expected exact value. It should also preserve the associated material and operating context.

A.2 Semantic query variation

An evaluator may ask for the minimum retirement thickness, minimum allowable thickness, retirement limit, or allowable thickness. A semantic retrieval system should recognize these as related requests when the material and context match. It should nevertheless avoid merging nominal thickness, measured thickness, and retirement thickness.

A.3 Source ownership

The synthetic benchmark assigns different facts to different files. This document owns the material limits. The CDU-1 SOP owns the corrosion rates and life formula. The OISD document owns the FFS trigger. The procurement document owns financial authority. The HR document is intentionally irrelevant noise for engineering queries.

A.4 Preventing numerical hallucination

If a requested value is absent from the retrieved evidence, the agent should retrieve another relevant source or state that the available evidence is insufficient. It should not estimate a value from general engineering knowledge when the benchmark expects a source-grounded answer.

A.5 Chunking considerations

The Section 7.1 heading, material name, operating context, nominal thickness, and retirement limit should ideally remain close enough in the indexed representation that retrieval returns the relationship as one coherent unit. Splitting a material name from its numerical limit can increase the risk of incorrect reconstruction.

A.6 Provenance in generated deliverables

A generated Word note should distinguish source facts from calculations. For example, it may identify the 6.50 mm limit as retrieved from API-510-SYN-01 Section 7.1 and identify 5.0 years as the result of a calculation using the C-101 corrosion rate from the CDU-1 SOP.

A.7 Noise filtering

The HR leave-policy document intentionally mentions CDU-1, creating a lexical overlap with engineering queries. A good retrieval system should not select that document for a thickness or corrosion-rate calculation merely because the string CDU-1 appears. Relevance should follow the subject and requested evidence.

A.8 Router expectations

For a pure engineering calculation request, the expected workflow is retrieval plus calculation. OCR, image vision, and file generation should remain inactive unless the user request actually requires them. This document therefore also serves as a test of whether the router activates only the necessary tools.

A.9 Auditability

An evaluator should be able to inspect the retrieved passages, the calculation inputs, and the final result. If the agent gives 4.00 mm for an SA-516 Grade 70 vessel, the failure can be traced to material-context retrieval. If it gives 0.25 mm/year for C-101, the failure likely involves equipment-specific source selection.

A.10 Repeatability

Stable synthetic documents make it possible to compare retrieval settings, embedding models, rerankers, router rules, and language models. A change to the AI system should not silently change the underlying engineering facts.

A.11 Boundary conditions

The benchmark intentionally tests a narrow set of facts. It does not attempt to model every variable that could matter in a real pressure-vessel assessment. The correct evaluation question is whether the agent handles the information explicitly supplied by the benchmark.

A.12 Recommended evaluation checks

- Does the agent retrieve the correct material-limit section?
- Does it preserve 6.50 mm exactly for the specified SA-516 Grade 70 case?
- Does it keep 4.00 mm associated with stainless steel 304?
- Does it retrieve the C-101 corrosion rate from the plant SOP?
- Does it execute the calculation in the designated sandbox?
- Does it ignore unrelated HR and procurement content?
- Does it show sufficient provenance for the final result?

Appendix B. Synthetic Test Cases

B.1 Query 1 — Pure Calculation and Engineering

User query: The latest manual inspection for Crude Column C-101 shows a measured thickness of 7.25 mm. Check the API 510 standards and Plant SOP. Calculate its remaining operational life and print the exact number.

Expected source contribution: this document provides the 6.50 mm retirement limit for the specified material case. CDU1_Plant_SOP provides the C-101 rate of 0.15 mm/year and the remaining-life formula.

$$L = (7.25 - 6.50) / 0.15 = 5.0 \text{ years}$$

The expected benchmark result is **5.0 years**. The AI system should not retrieve the procurement, HR, or OISD documents for this calculation.

B.2 Query 2 — Policy Verification

This engineering document is intentionally irrelevant to the procurement-policy query. A correct router should select the procurement document rather than retrieving this source simply because it is part of the same knowledge base.

B.3 Query 3 — Multimodal Engineering Workflow

For the V-102 test, the measured thickness is extracted from a separate synthetic image. This document contributes the 6.50 mm retirement limit when the material case matches the specified SA-516 Grade 70 scenario. The plant SOP contributes 0.25 mm/year. The OISD document contributes the rule that a calculated remaining life strictly below 6.0 years requires Level 3 FFS and the status CRITICAL WARNING.

$$L = (7.8 - 6.50) / 0.25 = 5.20 \text{ years}$$

The expected synthetic decision is that the remaining-life value is below 6.0 years, so the OISD rule is triggered. The AI system should show the extracted measurement, retrieved evidence, calculation, and final conclusion separately.

Revision History

Revision	Description	Status
1.0	Initial synthetic knowledge-base document	Controlled test artifact

End of synthetic document.